State Estimation and Filtering for Robotic Perception

AA273

Probability and RVs Review, Conditionals, Marginals,

Factorization, Bayes' Rule



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

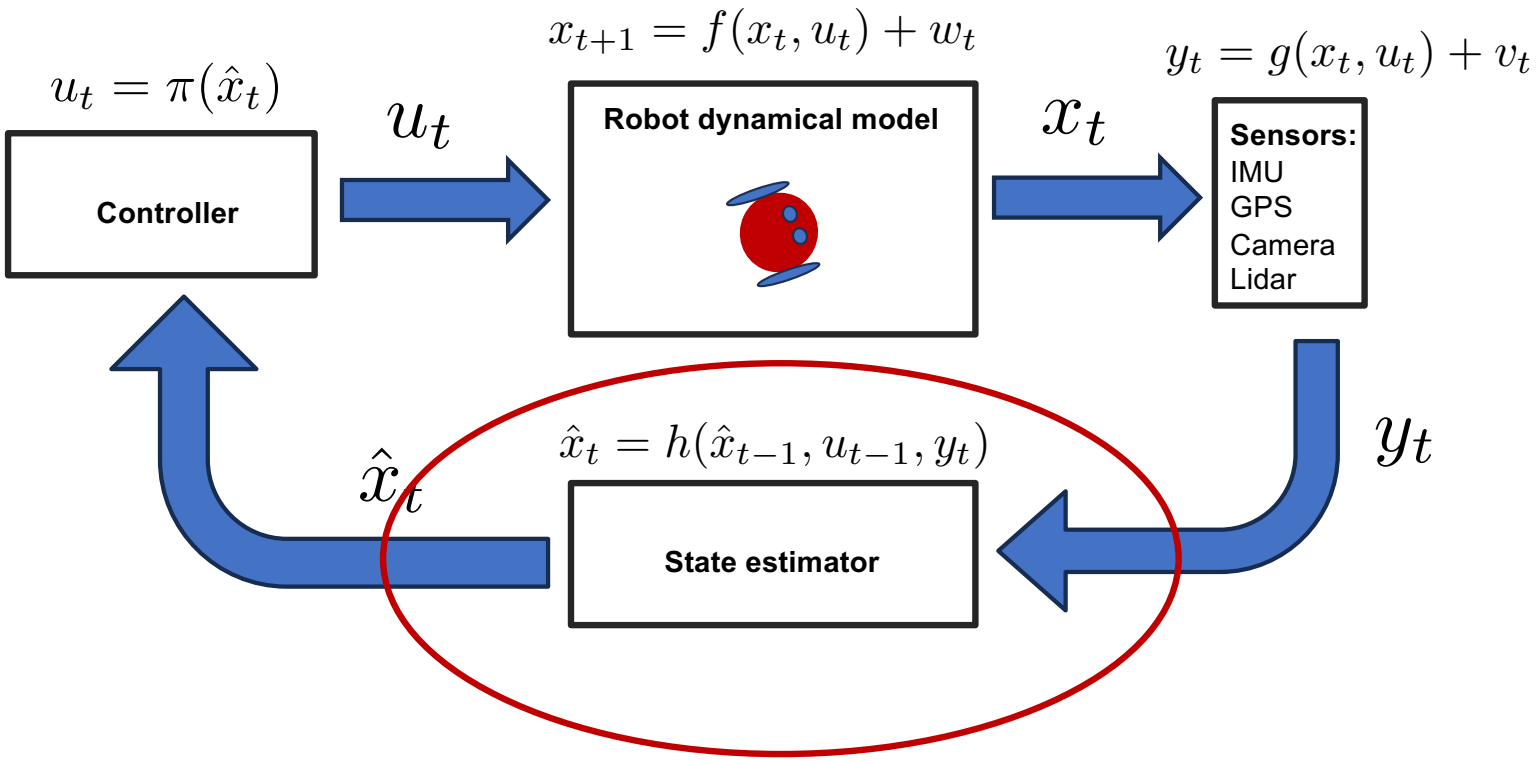
Announcements:

- HW 1 out tomorrow, due Fri, Jan 16
- Waitlisted students: Stick around for two weeks and keep up with the lecture and PSets. Check in with me early next week. You will likely get in.
- Project topic ideas will be posted to Canvas soon

Lecture 2:

- Probability and RV review
- Marginalization, factorization, rule of total probability
- Bayes' rule
- Robot localization example

State Estimation Overview



Overall Problem Statement (deterministic)

Given model:

Dynamics model: $f(\cdot, \cdot)$ Process noise distribution: $W_t \sim D(w_t)$

Sensor model: $g(\cdot, \cdot)$ Sensor noise distribution: $V_t \sim D(v_t)$

Given data:

Measurement history: $(y_1, \ldots, y_t)$ Control history: $(u_0, \ldots, u_{t-1})$

Find:

A state estimate: $\hat{x}_t$

Overall Problem Statement (probabilistic)

Given model:

Dynamics model: $f(\cdot, \cdot)$ Process noise distribution: $W_t \sim D(w_t)$

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Given data:

Measurement history: $(y_1, \ldots, y_t)$ Control history: $(u_0, \ldots, u_{t-1})$

Find:

The probability distribution over the state: $b(x_t) := p(x_t \mid y_{1:t}, u_{0:t-1})$

$b(x_t)$ is called the belief, or the Bayesian posterior

Linear-Gaussian Systems and the Kalman Filter

For linear state space models with Gaussian white noise:

Dynamics model: $x_{t+1} = A_t x_t + B_t u_t + w_t$

Process noise: $w_t \sim GWN(0, Q_t)$

Sensor model: $y_t = C_t x_t + D_t u_t + v_t$

Sensor noise: $v_t \sim GWN(0, R_t)$

…and other technical conditions,

the Bayesian posterior is Gaussian: $\boxed{b(x_t) = \mathcal{N}(\mu_{t|t}, \Sigma_{t|t})}$

where the mean $\mu_{t|t}$ and covariance $\Sigma_{t|t}$ are found very efficiently with an algorithm called the **Kalman Filter**.

We'll see the details and precisely define notation in the coming lectures…

Non-Linear, Non-Gaussian Filters

For linear state space models with Gaussian white noise:

Dynamics model: $x_{t+1} = f(x_t, u_t) + w_t$

Process noise: $w_t \sim D_w$

Sensor model: $y_t = g(x_t, u_t) + v_t$

Sensor noise: $v_t \sim D_v$

EFK: Linearize dynamics and measurement model about current estimate. Maintains Gaussian posterior approximation like KF: $(\mu_{t|t}, \Sigma_{t|t})$

UFK: Similar philosophy to EKF, but linearize performed with special sample points called sigma-points. Also maintains approximate Gaussian posterior: $(\mu_{t|t}, \Sigma_{t|t})$

PF: Uses a weighted sample set of states to represent posterior. Propagates samples stochastically and updates with importance weighting and statistical re-sampling.

Probability and Random Variables Review

- Key idea: quantities such as sensor measurements, states of a robot, and its environment are modeled as random variables (RVs)
- Discrete RV: the space of all the values that a random variable *X* can take on is *discrete*; characterized by probability mass function (pmf)

$$p(x) := Pr(X = x)$$

Random variable — Specific value

$$(i) \quad \sum_x p(x) = 1 \qquad (ii) \quad 0 \leq p(x) \leq 1 \quad \forall x$$

- Continuous RV: the space of all the values that a random variable *X* can take on is *continuous;* characterized by probability density function (pdf)

$$p(x) \text{ s.t. } Pr(X \in S) = \int_{x \in S} p(x)\, dx \quad \forall S \subset D_x$$

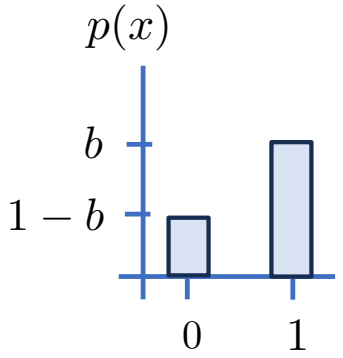
$$(i) \quad \int_{x \in D_x} p(x)\, dx = 1 \qquad (ii) \quad p(x) \geq 0 \quad \forall x \in D_x$$

Distribution Families

Analytical distributions with specific instances defined by parameters.

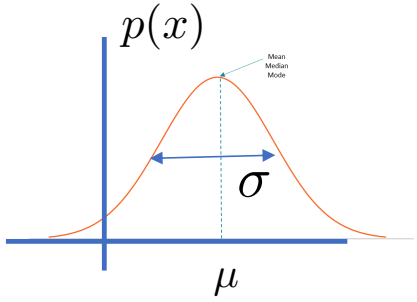
- E.g. Bernoulli RV (coin toss):

$$x \in \{0, 1\} \qquad p(x) = \begin{cases} b & \text{for } x = 1 \\ 1 - b & \text{for } x = 0 \end{cases}$$



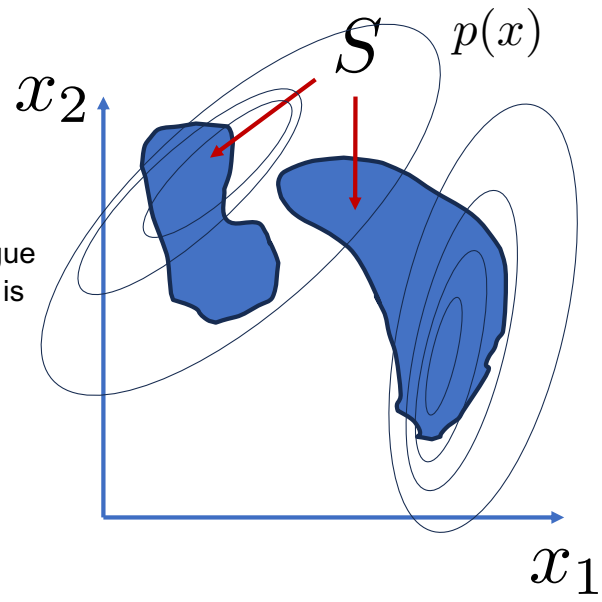
- E.g. Gaussian Distributed RV:

$$x \in \mathbb{R} \qquad p(x) = \frac{1}{\sqrt{(2\pi)}\sigma} e^{\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$$



Same for jointly dist. RVs, and Random Vectors

$$X = [X_1 \quad X_2 \quad \cdots \quad X_n]^T \in \mathbb{R}^n$$



Joint dist. defined as:

Technicality: S must be Lebesgue Integrable to ensure the integral is well defined

$$P(X \in S) = \int_{x \in S} p(x)\, dx \quad \forall S \subset \mathbb{R}^n$$

Multivariate integral

$$(i) \quad \int_{\mathbb{R}^n} p(x)\, dx = 1 \qquad (ii) \quad p(x) \geq 0 \quad \forall x \in \mathbb{R}^n$$

Joint distribution, independence, and conditioning

- Joint distribution of two random vectors *X* and *Y* is denoted as

$$p(x, y) = p(z) \qquad z = (x, y)$$

- *X* and *Y* are independent iff

$$p(x, y) = p(x)p(y)$$

- Suppose we know that $Y = y$ (with $p(y) > 0$); conditioned on this fact, the probability that the X's value is x is given by

$$p(x \mid y) := \frac{p(x, y)}{p(y)}$$

Conditional probability

Note: if *X* and *Y* are independent

$$p(x \mid y) = p(x)$$
$$p(y \mid x) = p(y)$$

Law of total probability

- For discrete RVs:

$$p(x) = \sum_y p(x, y) = \sum_y p(x \mid y)p(y)$$

- For continuous RVs:

$$p(x) = \int p(x, y)dy = \int p(x \mid y)p(y)dy$$

- Note: if $p(y) = 0$, define the product $p(x \mid y)p(y) = 0$

Bayes' rule

- Key relation between $p(x \mid y)$ and its "inverse," $p(y \mid x)$
- For discrete RVs:

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)} = \frac{p(y \mid x)p(x)}{\sum_{x'} p(y \mid x')p(x')}$$

- For continuous RVs:

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)} = \frac{p(y \mid x)p(x)}{\int p(y \mid x')p(x')\,dx'}$$

Bayes' rule and probabilistic inference

- Assume *x* is a quantity we would like to infer from *y*
- Bayes rule allows us to do so through the inverse probability, which specifies the probability of data *y* assuming that *x* was the cause

Posterior probability distribution

Prior probability distribution

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{\int p(y \mid x')p(x')\,dx'}$$

Data

Normalizer, does not depend on $x := \eta^{-1}$

- Notational simplification

$$p(x \mid y) = \eta\, p(y \mid x)p(x)$$